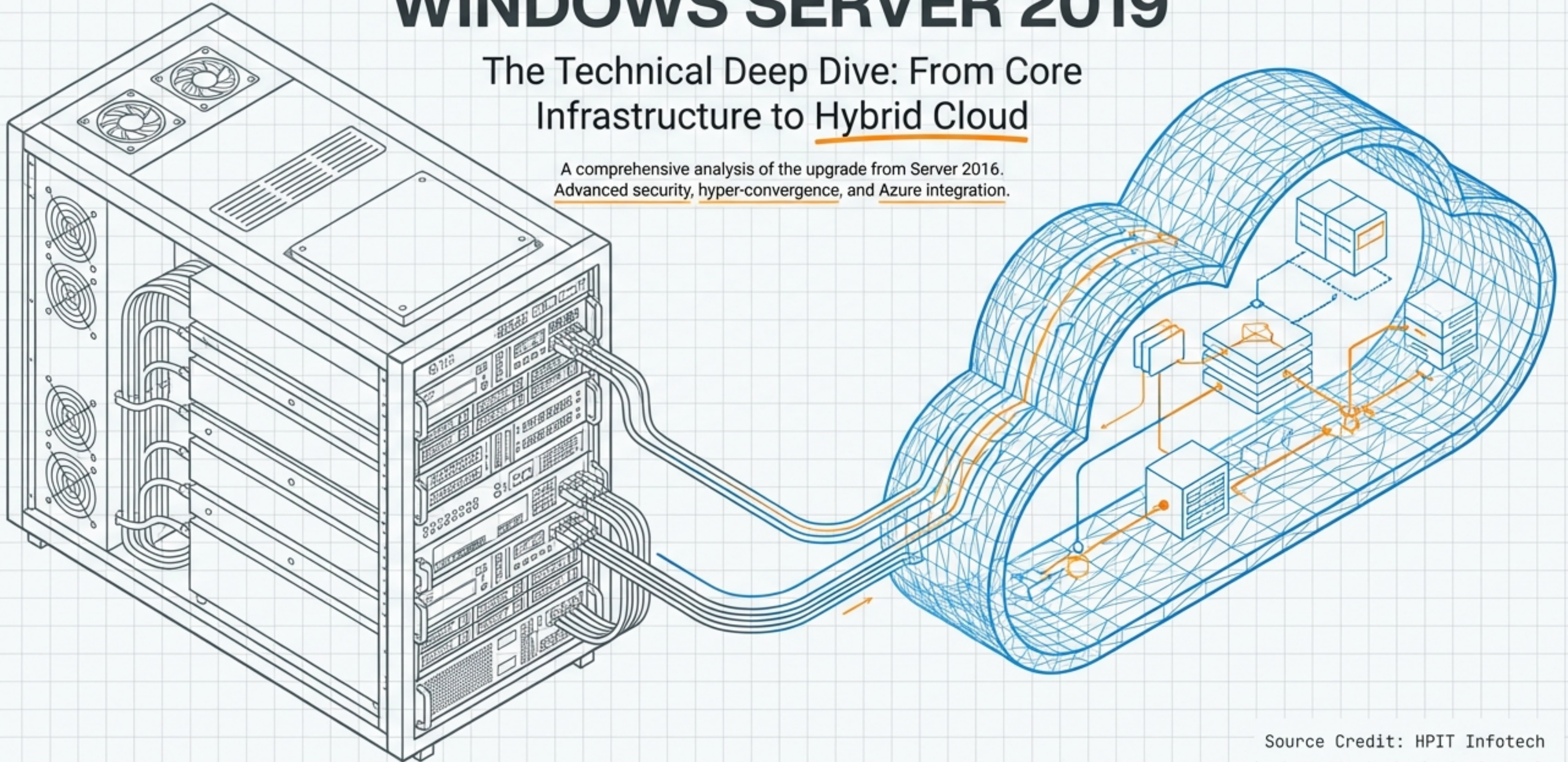


Precision Engineering Blueprint

WINDOWS SERVER 2019

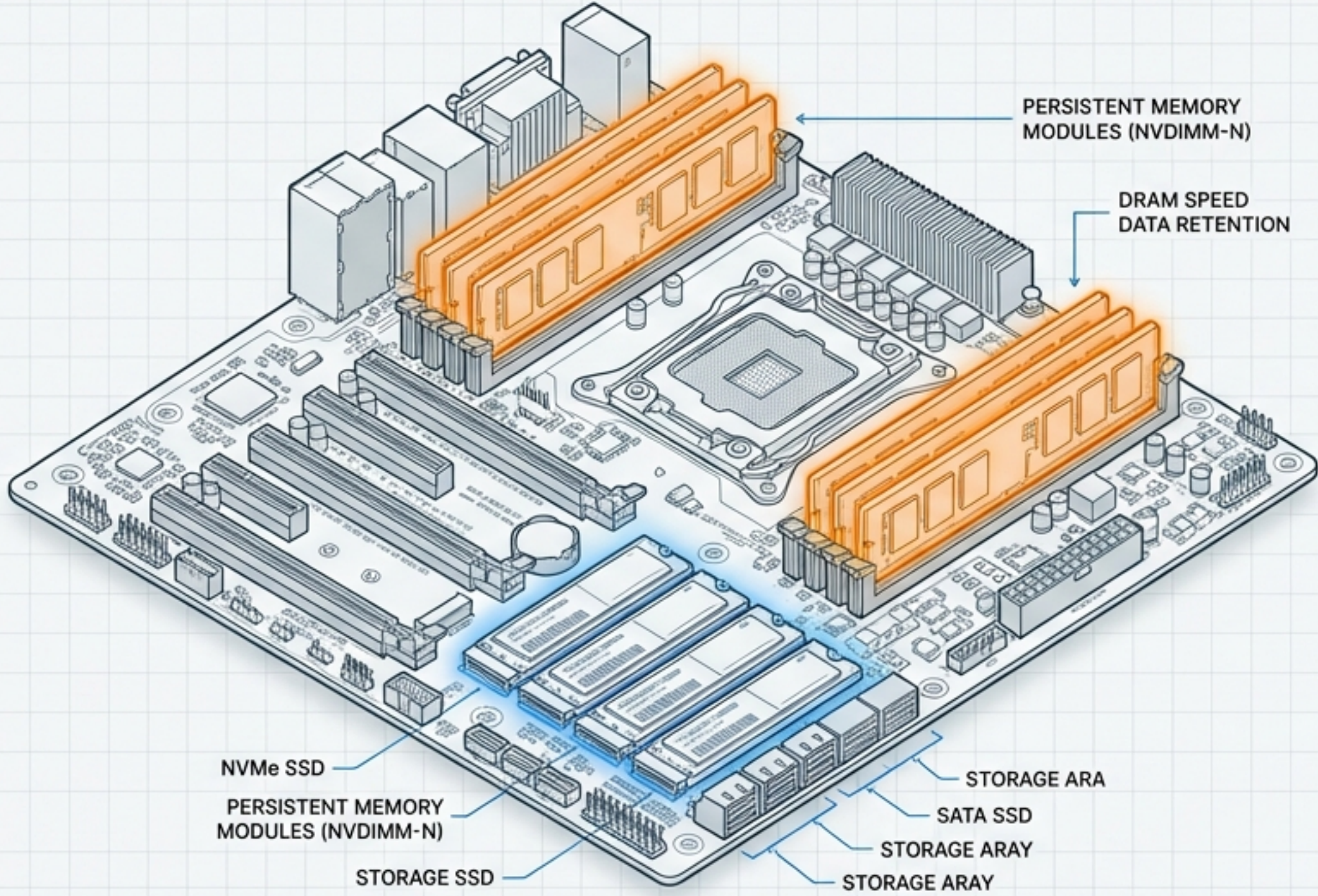
The Technical Deep Dive: From Core Infrastructure to Hybrid Cloud

A comprehensive analysis of the upgrade from Server 2016.
Advanced security, hyper-convergence, and Azure integration.



Source Credit: HPIT Infotech

HARDWARE & STORAGE EVOLUTION



Persistent Memory (NVDIMM-N)

Supports NVDIMM-N technology, delivering DRAM speeds with full data persistence. Unlike traditional DRAM, data is retained during power cycles.

Storage Migration Service

A fully automated tool designed to inventory and migrate legacy servers to new targets.

TECHNICAL SPECS



Capacity Limit:
32 Terabytes (Virtual TB)



Memory Type:
Persistent Class



Target:
High-Performance SQL/Hyper-V

THE INTELLIGENCE LAYER: SYSTEM INSIGHTS

Native Predictive Analytics

Local system data analysis using built-in AI and Machine Learning models. Provides performance forecasting without requiring external cloud agents.



TECHNICAL SPECS



Tool:
System
Insights

Engine:
Local
Machine
Learning



Dependency:
None
(Offline
Capable)

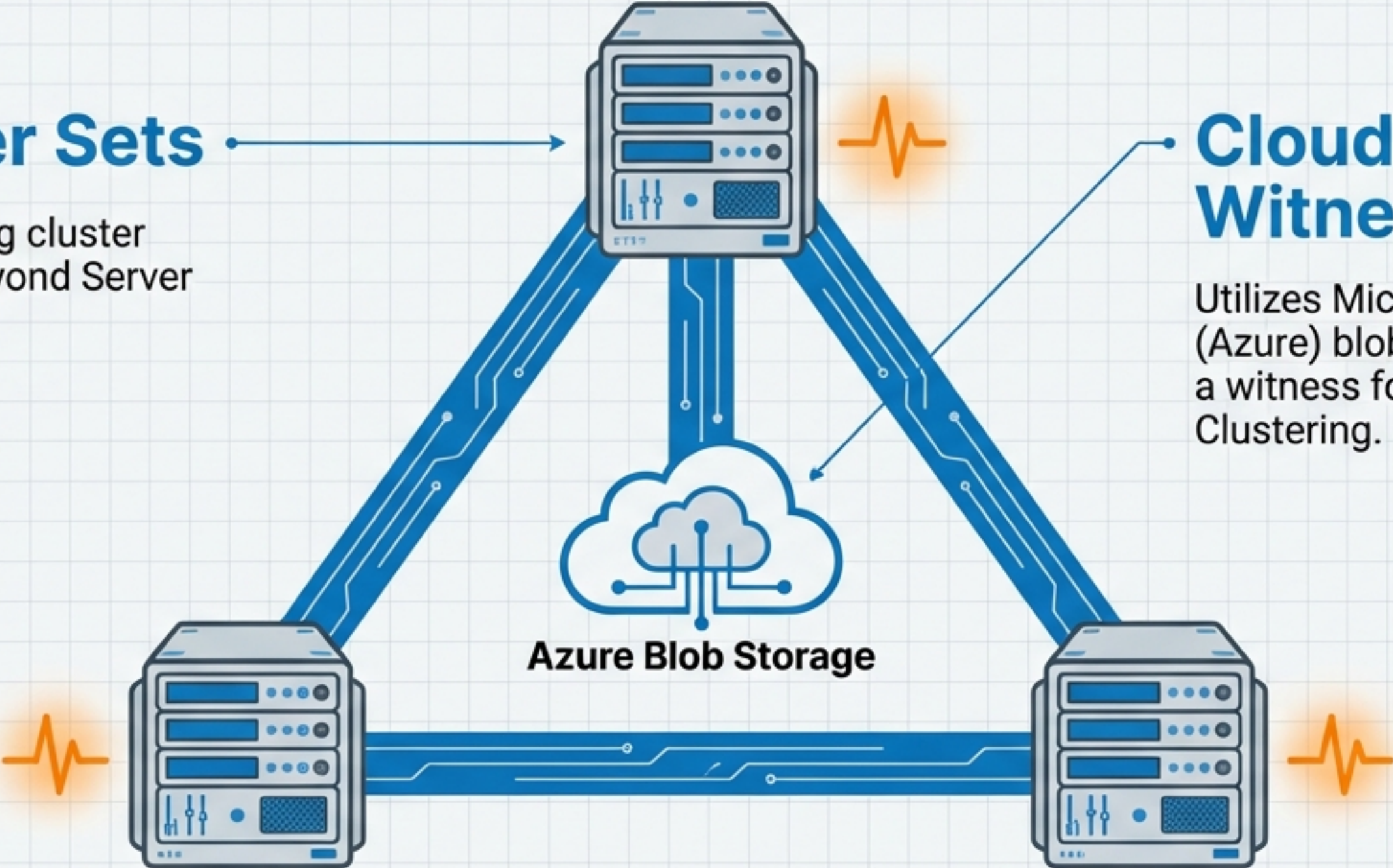
PROJECT:
NEXT-GEN SERVER
INFRASTRUCTURE

DRAWING NO:
HSW-001-REV A
DATE: OCTOBER 26, 2023

HCI & CLUSTERING RESILIENCE

Cluster Sets

Hyperscaling cluster services beyond Server 2016 limits.



Cloud Witness

Utilizes Microsoft Cloud (Azure) blob storage as a witness for Failover Clustering.

TECHNICAL SPECS



Metric:
**Real-time
Status
Checkers**

Monitoring:
CPU & IOPS

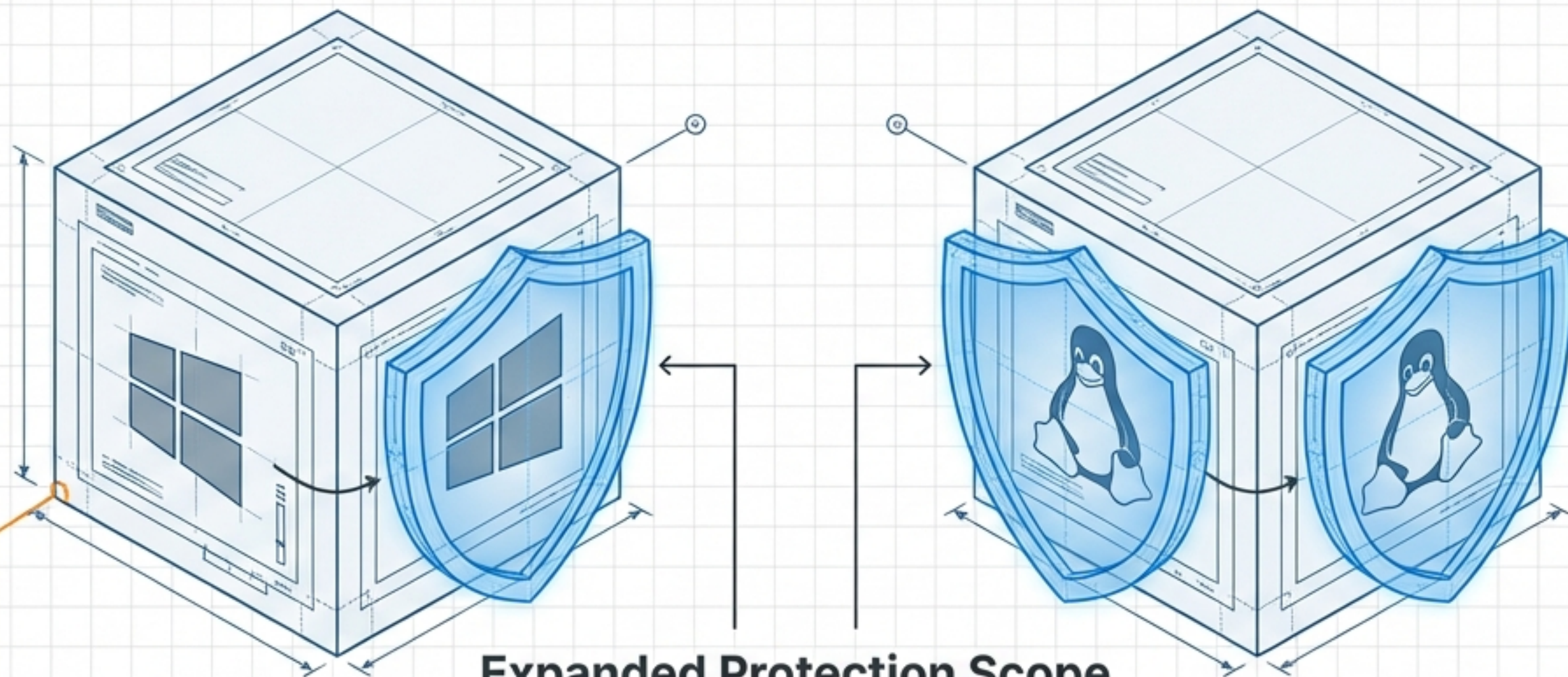
Latency:
**Real-time
Storage
Tracking**

Architecture:
**Hyper-
Converged**

PROJECT:
NEXT-GEN SERVER
INFRASTRUCTURE

DRAWING NO:
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DATE: OCTOBER 26, 2023

VIRTUALIZATION: SHIELDED VMs



Expanded Protection Scope

Shielded VM support extended to Linux Operating Systems.
(Previously Windows-only in Server 2016).



HGS Offline Mode

Allows Shielded VMs to power on even if the Host Guardian Service is unreachable.

TECHNICAL SPECS

 **Support:**
Windows & Linux

 **Optimization:**
Smaller Container
Image Sizes

 **Security:**
Encryption at
Rest & Flight

PROJECT:
NEXT-GEN SERVER INFRASTRUCTURE

DRAWING NO:
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OCTOBER 26, 2023

SECURITY PERIMETER: DEFENDER ATP

TECHNICAL SPECS

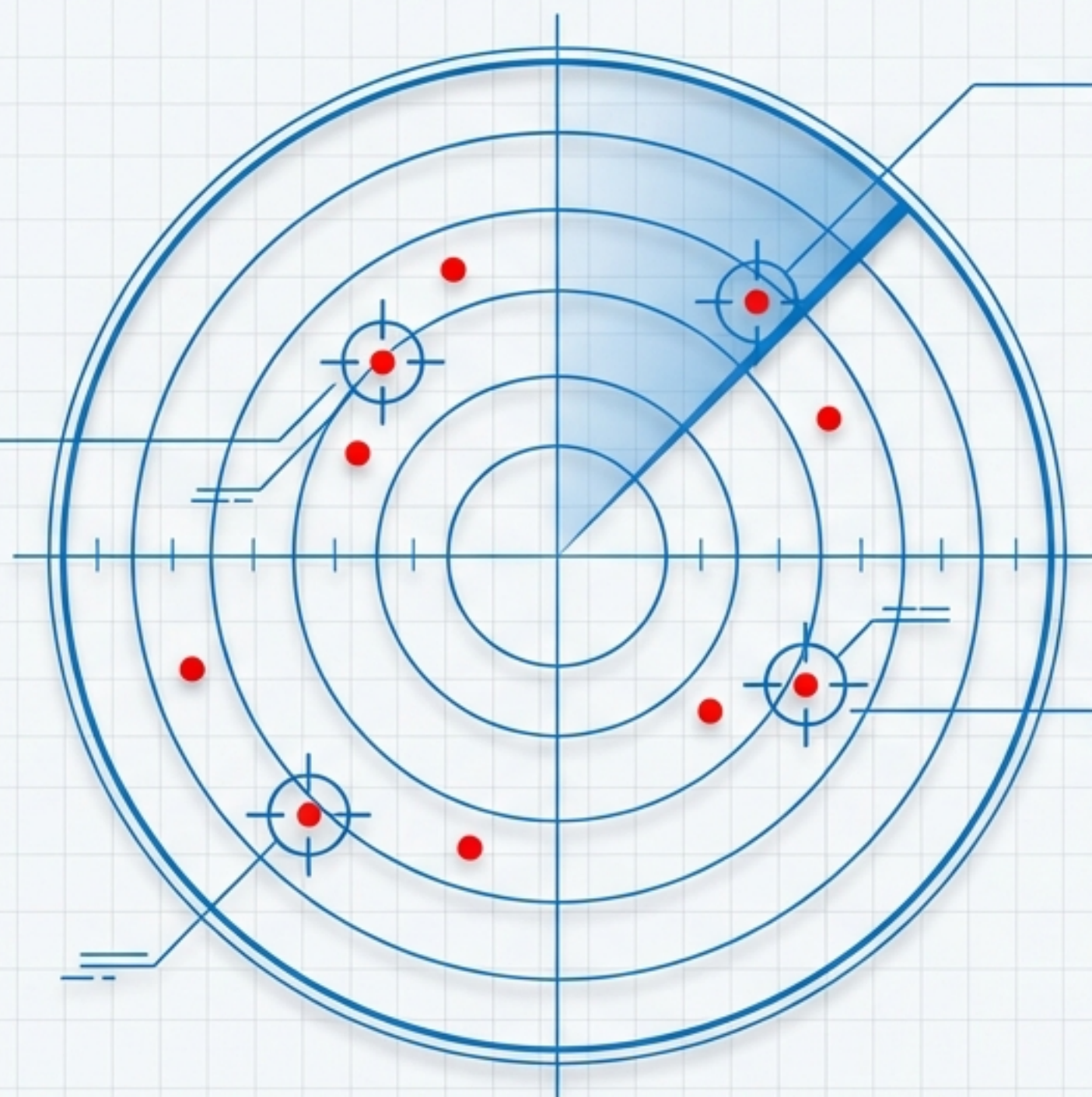
Suite:
Windows Defender
ATP

Methodology:
Prevent, Detect,
Respond

Scope:
Kernel & User Mode

Preventive Protection

Platform sensors
designed to block
attacks before
execution.



Attack Detection

Behavioral monitoring
to identify breach
attempts.

Zero-Day Exploits

Handling unknown
vulnerabilities and
sophisticated threats.

PROJECT: NEXT-GEN SERVER INFRASTRUCTURE

DRAWING NO: HSW-001-REV A

DATE: OCTOBER 26, 2023

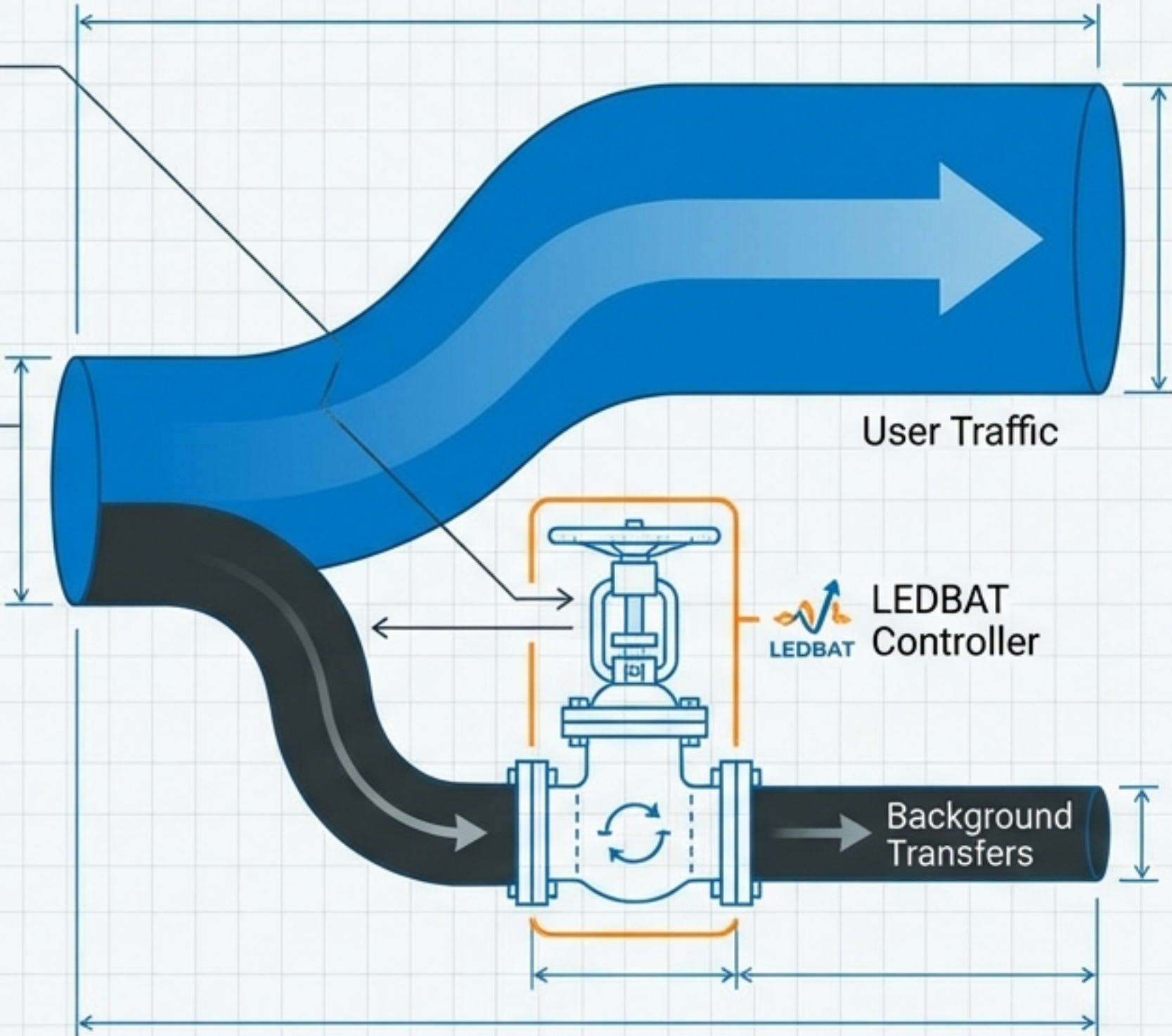
NETWORKING: THROUGHPUT & FLOW

LEDBAT Protocol

Low Extra Delay Background Transfer. Scavenges available bandwidth for updates without impacting user performance.

SDN Gateway Performance

Software Defined Networking Gateway updated for high-demand environments.



TECHNICAL SPECS

📈 **Performance:**
3x Faster than
Server 2016

⚙️ **Function:**
Network
Congestion
Control

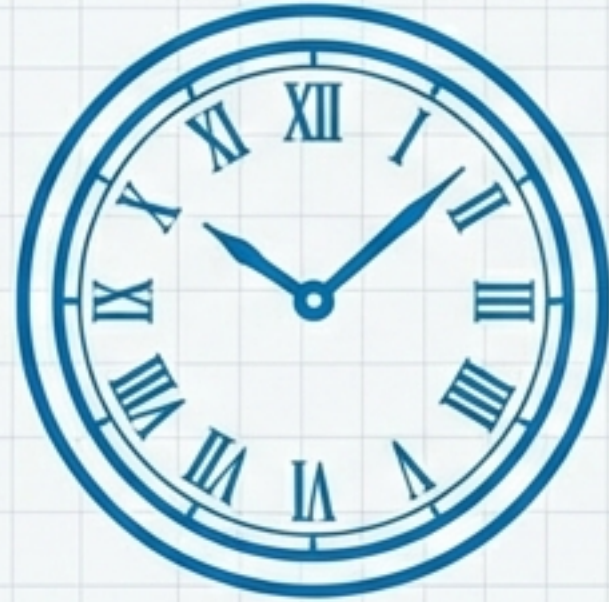
🔒 **Tunnels:**
GRE & IPSec
Optimized

PROJECT: NEXT-GEN SERVER INFRASTRUCTURE

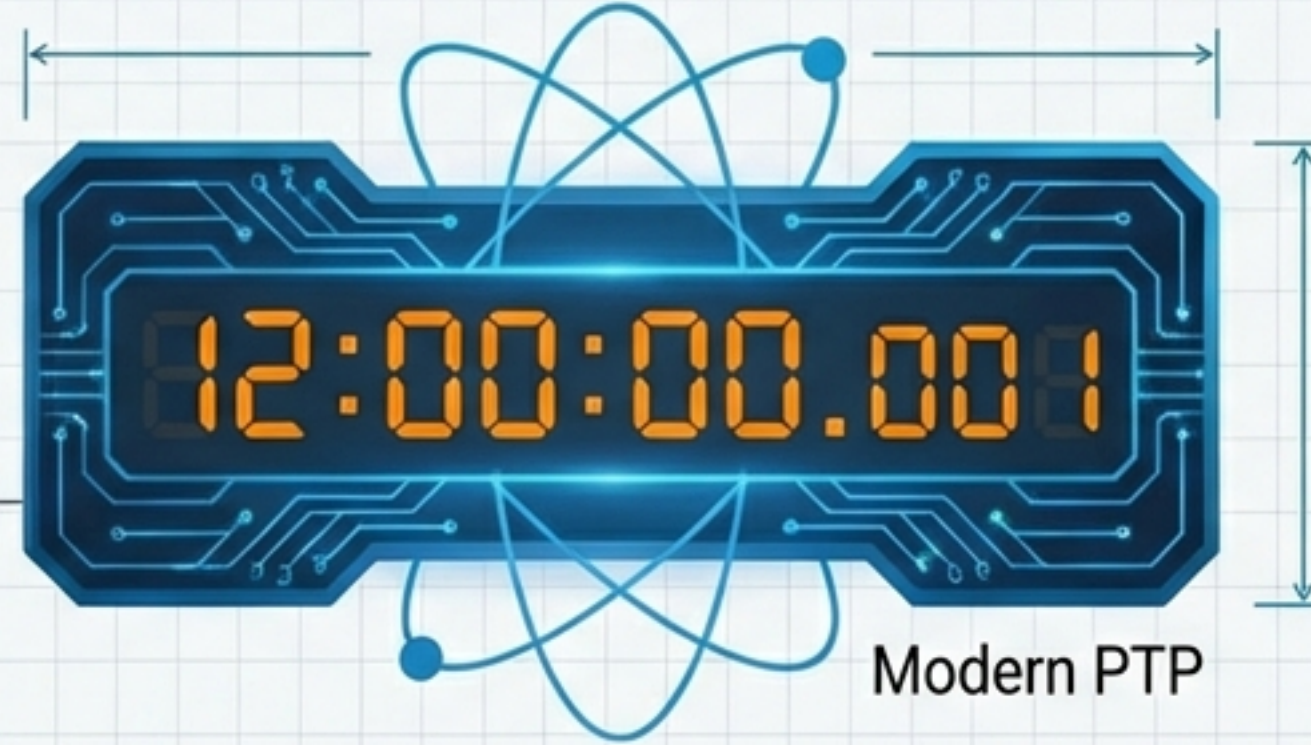
DRAWING NO: HSW-001-REV A

DATE: OCTOBER 26, 2023

NETWORKING: PRECISION & PROTOCOL



Legacy NTP
Roboto



Precision Time Protocol (PTP)

Replaces and augments NTP. delivers **microsecond-level accuracy** essential for financial compliance and regulated industries.

TECHNICAL SPECS

Standard:
PTP (Precision
Time Protocol)

Accuracy:
Sub-millisecond

Addressing:
IPv6 / Dual Stack

Addressing Standards

Full support for IPv6 addressing schemes and Dual Stack (IPv4/IPv6) simultaneous operation.

PROJECT: NEXT-GEN SERVER INFRASTRUCTURE

DRAWING NO: HSW-001-REV A

DATE: OCTOBER 26, 2023

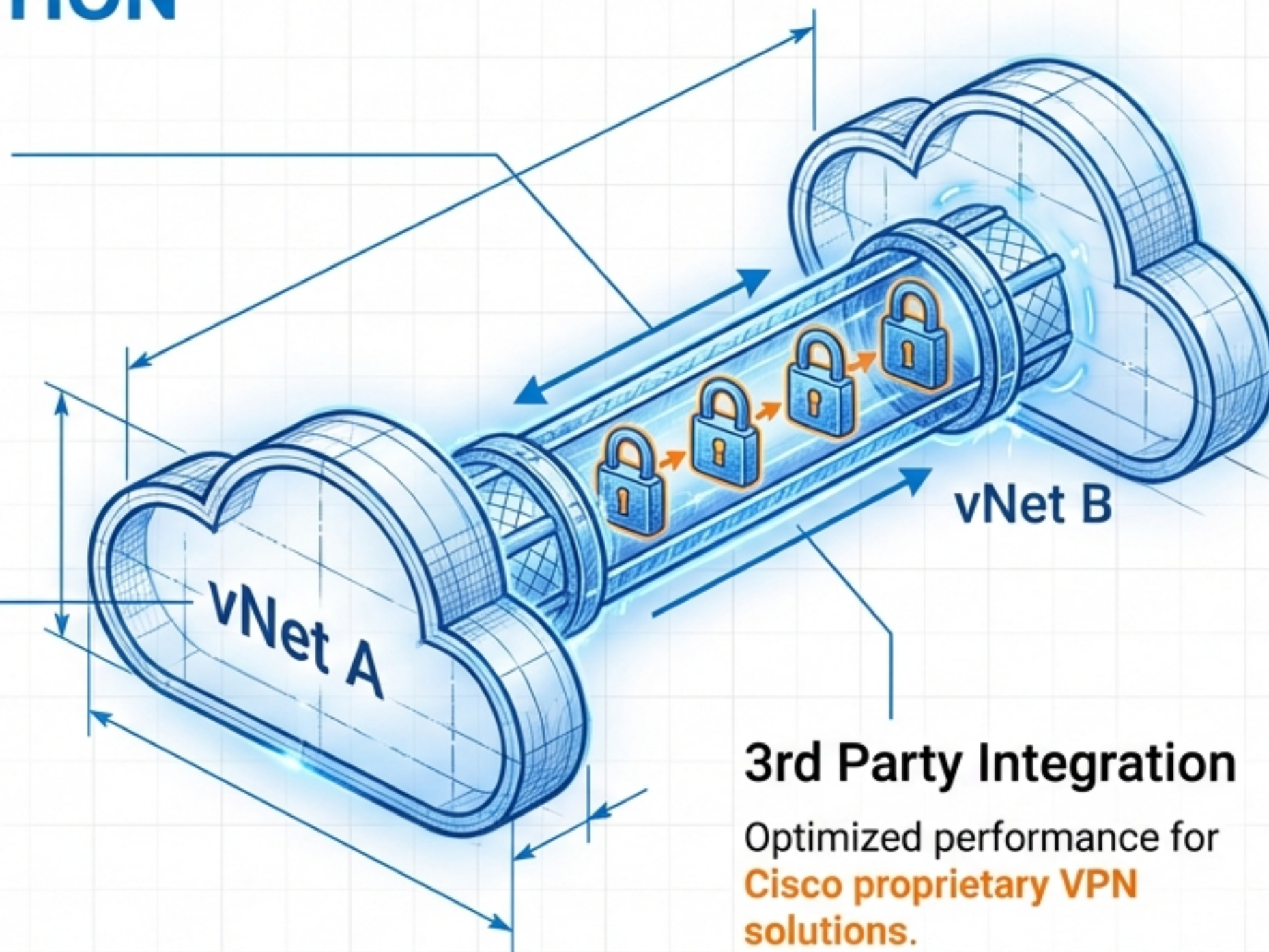
NETWORKING: CONNECTIVITY & ENCRYPTION

Virtual Network Peering

Seamlessly connects vNets, routing data between them as a unified network.

Encrypted Subnets

Prevents internal sniffing by applying encryption within the virtual network subnets.



3rd Party Integration

Optimized performance for Cisco proprietary VPN solutions.

TECHNICAL SPECS

Security:
Virtual Network
Encryption

Topology:
vNet Peering

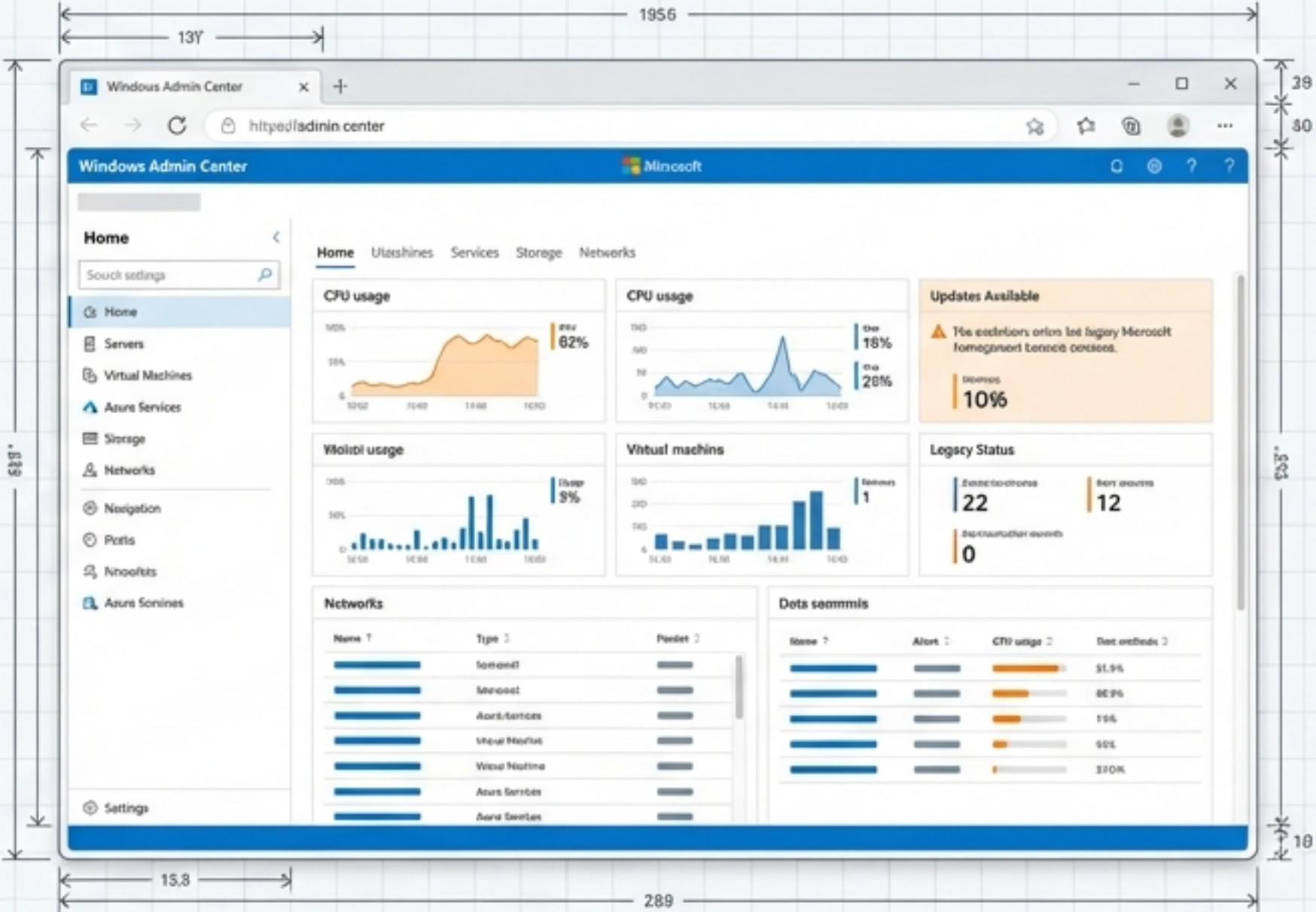
Integration:
Cisco VPN Support

MANAGEMENT: WINDOWS ADMIN CENTER

Modern Management

The evolution from the legacy Microsoft Management Console (MMC) to a comprehensive browser-based platform.

- Centralized control of Hybrid resources.
- Native Gateway to Azure integration.
- Deployed locally, accessed via browser.



TECHNICAL SPECS

Platform:
Browser-Based (HTML5)

Scope:
Hybrid (On-Prem + Cloud)

Legacy Status:
Replaces MMC

TECHNICAL SPECIFICATION SUMMARY

1. VIRTUALIZATION

- ✓ **Shielded VMs:**
Now supports Linux
- ✓ **HGS:** Offline Mode supported
- ✓ **Containers:**
Reduced image sizes

2. NETWORKING

- ✓ **Protocol:**
PTP replaces NTP
- ✓ **Congestion:** **LEDBAT**
Background Transfer
- ✓ **Gateway:** SDN **3x**
Faster than 2016
- ✓ **Addressing:** **IPv6**
& Dual Stack

3. STORAGE & SYSTEM

- ✓ **Capacity:** **32TB**
Virtual TB Limit
- ✓ **Memory:** Persistent
Memory (NVDIMM-N)
- ✓ **Migration:** Storage
Migration Service
- ✓ **Analytics:** System
Insights (**4 Models**)

4. SECURITY

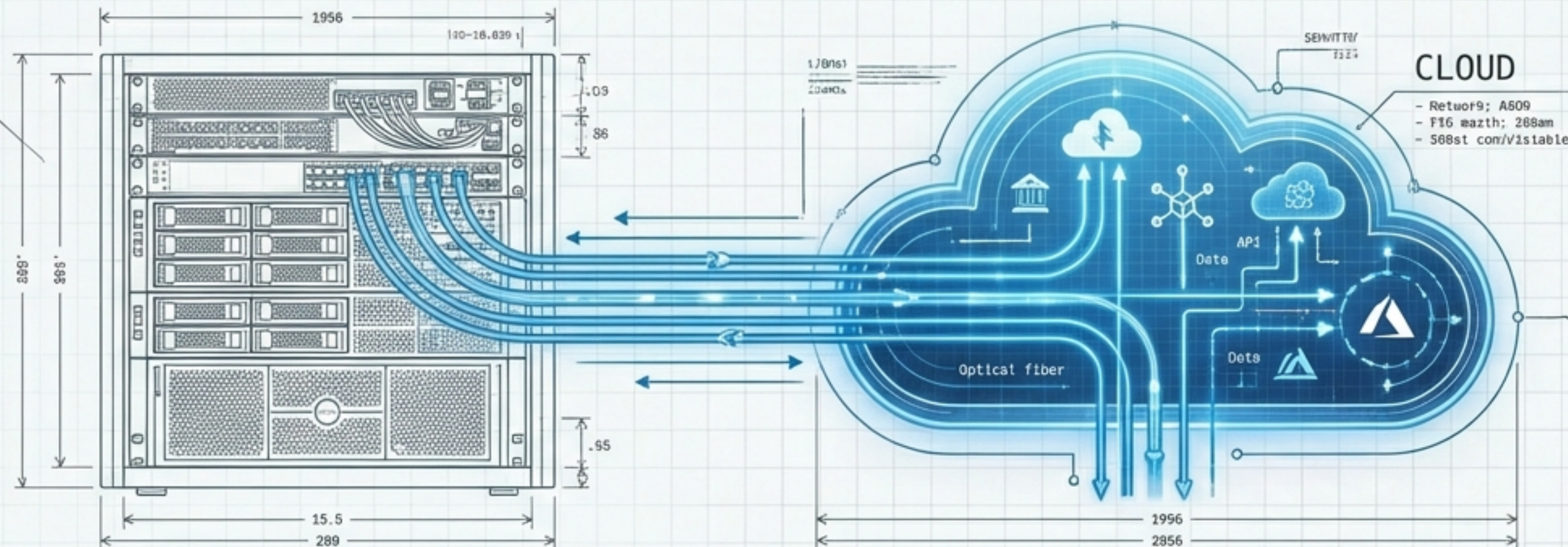
- ✓ **Defender ATP:**
Zero-day protection
- ✓ **Network:**
Encrypted Subnets

THE HYBRID FUTURE IS HERE



IRON

- Service: B90N
- 786 weight: 236mm
- 1250at con-711s201t



SECURITY

From Passive to Predictive.
Defending against **Zero-day threats** with **ATP**.

INFRASTRUCTURE

From Manual to Automated.
Self-optimizing with **System Insights** and **Migration Services**.

CONNECTIVITY

From Standard to Precise.
Time-accurate **PTP** and **congestion-free LEDBAT**.

Windows Server 2019: The bridge between on-premises stability and Azure innovation.